

CS 3151
Final Project

Software Engineering

Project: Cognitive Health Monitoring System

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Executive Summary

The purpose of this document is to specify the comprehensive software requirements for the **Cognitive Health Monitoring System**. This system is intended to passively monitor the cognitive health of senior citizens by analyzing their longitudinal social media text patterns, thereby detecting early semantic indicators of cognitive decline such as dementia and Alzheimer's disease.

Scope and Market Need: Currently, dementia is often diagnosed late due to the stress, cost, and infrequency of clinical memory tests. Our market study indicates a massive demand for non-invasive, continuous screening tools. The system provides secure API integration with social networks, automated 100% PII anonymization via Named Entity Recognition (NER), and longitudinal risk scoring using a fine-tuned ClinicalBERT AI model. It is designed for seniors, caregivers, clinical neurologists, and IT administrators, deployed as a secure cloud-based web platform utilizing big data architectures.

Key Objectives:

- Improve the efficiency and timeline of early dementia detection using retrospective digital data.
- Provide standardized, objective behavioral data to clinical neurologists via high-speed dashboards.
- Ensure secure, scalable, and fully anonymized data processing that complies with global medical privacy laws.

Benefits & Risks: This system will radically reduce the diagnostic delay for neurodegenerative diseases and minimize patient anxiety by eliminating active testing. However, it relies heavily on the availability of the Facebook Graph API and requires strict adherence to evolving data privacy regulations regarding AI processing. Ultimately, this system delivers a robust, semantically-aware platform that bridges the gap between academic Natural Language Processing research and practical, preventive patient care.

1. Introduction

This chapter introduces the foundational concepts, clinical goals, and architectural boundaries of the Cognitive Health Monitoring System. It explores the existing academic literature surrounding the use of artificial intelligence in detecting cognitive decline and outlines the competitive landscape of the digital health market.

1.1 Market Study and Feasibility

The Global Healthcare Crisis: The global population is aging rapidly, and neurodegenerative diseases—particularly Alzheimer's and dementia—represent one of the most critical and expensive challenges in modern healthcare. According to recent epidemiological studies, over 50 million people currently live with dementia globally, a number projected to triple by 2050. The financial burden of late-stage dementia care places immense pressure on healthcare systems, caregivers, and families. Early detection is widely considered the most effective way to slow disease progression and manage long-term care costs.

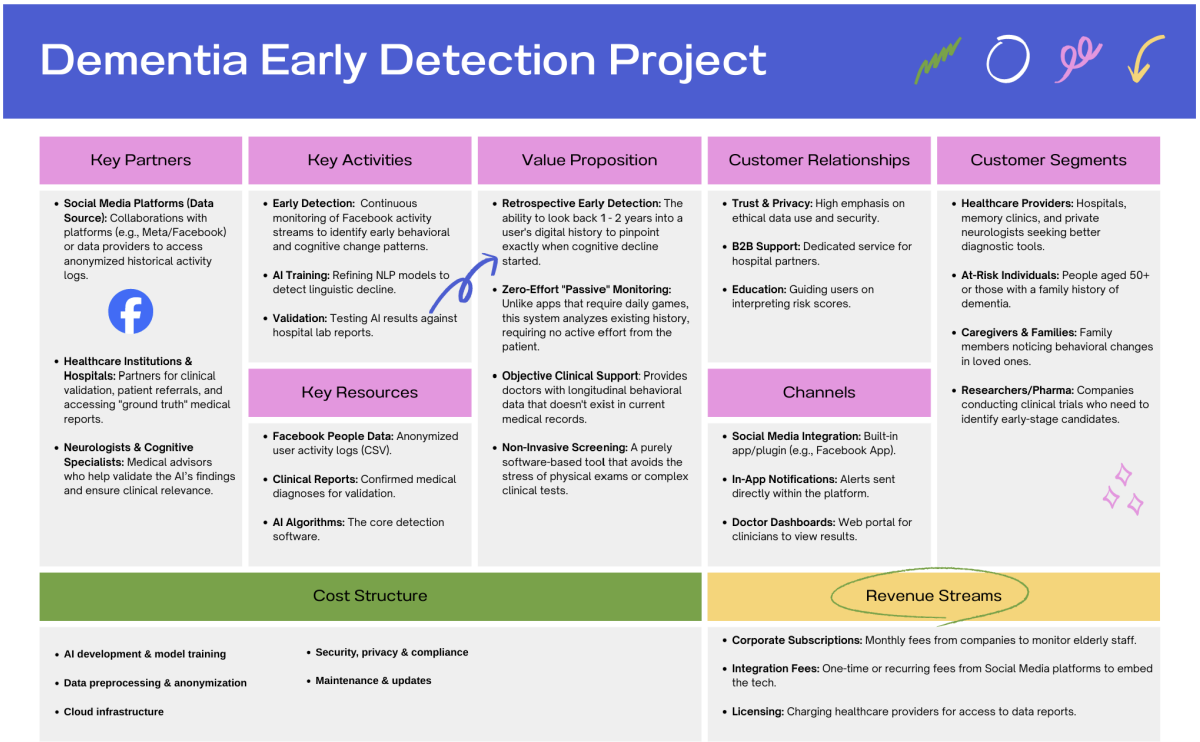
The Diagnostic Gap: Traditionally, cognitive decline is diagnosed through episodic clinical assessments, such as the MMSE (Mini-Mental State Examination). However, these tests suffer from severe limitations. They are administered infrequently (often years apart), are expensive to process, and cause significant anxiety for the patient (often leading to "white-coat syndrome" which skews results). Consequently, diagnoses are usually made only after noticeable, irreversible cognitive impairment has already occurred, missing the crucial early-intervention window.

The "Zero-Effort" Market Opportunity: Our market analysis reveals a booming digital health sector saturated with "active" brain-training applications (e.g., Lumosity, Peak). While popular, these applications suffer from extremely low long-term retention rates among seniors, as they require active daily effort and cause "app fatigue." Conversely, individuals aged 60 and older represent the fastest-growing demographic on social media platforms like Facebook. This creates a massive, untapped market for **"zero-effort" passive monitoring**. By utilizing data the user is *already generating* in their daily life, the system achieves near 100% long-term compliance without requiring the user to change their habits or take stressful tests.

Technical and Economic Feasibility: The proposed software system is highly feasible from both a technical and economic standpoint. Rather than requiring millions of dollars in RD to train foundation neural networks from scratch, the system leverages open-source, pre-trained Transformer models (ClinicalBERT) fine-tuned for semantic degradation. Furthermore, because it relies on API-based data ingestion rather than proprietary biometric hardware (like EEG headsets or smartwatches), the deployment costs are exceptionally low. The primary risk is data privacy, which is heavily mitigated by deploying an automated Named Entity Recognition (NER) pipeline at the edge to strip all Personally Identifiable Information (PII) before it ever reaches the permanent database. Ultimately, the core objective of this early intervention and continuous monitoring is to preserve cognitive longevity—extending the patient's mental "healthspan" so they can maintain their independence and quality of life for as long as possible.

1.1.1 Business Model Canvas

The following canvas maps the economic and strategic viability of the proposed system based on the market study findings.



1.2 Literature Review and Related Work

The architectural and algorithmic foundations of this system are deeply rooted in recent advancements in Natural Language Processing (NLP) and Big Data infrastructure. A systematic review by Soni et al. highlights the vast opportunities of applying NLP to dementia detection, noting that semantic degradation is one of the earliest measurable symptoms of the disease. Building upon this, Peled-Cohen Reichart demonstrated the efficacy of Large Language Models (LLMs) in accurately rendering judgments of cognitive impairment based purely on unstructured language inputs. By utilizing a fine-tuned ClinicalBERT model, our system replicates this explainable AI approach to detect subtle fluctuations in vocabulary shrinkage and syntactic confusion.

However, translating these theoretical AI capabilities into a scalable clinical tool requires massive computational infrastructure. Research into healthcare analytics has proven that traditional monolithic databases fail under the weight of unstructured clinical text. According to extensive studies on Big Data processing systems in clinical research, integrating Apache Spark with Hadoop data lakes is essential for handling high-throughput, multimodal data pipelines. Our system adopts this exact distributed architecture, utilizing Spark for batch-processing up to 10,000 text logs per hour and Hadoop for secure, long-term storage, successfully fusing advanced NLP with enterprise-grade scalability.

1.3 Definitions, Acronyms, Abbreviations

To ensure all stakeholders have a clear understanding of the technical and medical jargon used in this document, the following definitions are provided in Table 1.1.

Term	Definition
PII	Personally Identifiable Information. Data that identifies a specific individual (names, locations).
NLP	Natural Language Processing. AI focused on interpreting human language.
ClinicalBERT	A pre-trained Transformer AI model specifically fine-tuned to understand clinical text variations.
NER	Named Entity Recognition. An algorithm that locates and strips personal entities from raw text.
Digital Biomarker	Objective behavioral data (like vocabulary use) collected via digital means to measure health.
HIPAA	Health Insurance Portability and Accountability Act. A federal law mandating strict privacy and security standards for protecting medical information.
OAuth 2.0	Open Authorization 2.0. The industry-standard protocol for secure authorization, allowing the system to access Facebook data without storing user passwords.

Table 1.1: System Definitions and Acronyms

1.4 Competitor Analysis

Our system distinguishes itself from existing real-world applications by requiring absolute zero effort from the user while providing long-term, medically actionable data to clinicians.

Competitor App	Passive Monitoring	Longitudinal Trends	Data Anonymization
Lumosity [5] (Brain training)	No (Requires active daily gaming)	Limited to app scores	Generic user privacy
Winterlight Labs [6] (Speech analysis)	No (Requires clinical recording)	Yes (High clinical accuracy)	Yes (Medical grade)
Mindstrong [7] (Keyboard tracking)	Yes (Tracks typing habits)	Yes	Yes
Our AI System (Social Media NLP)	**Yes (Zero user effort)**	**Yes (5 years of history)**	**Yes (100% NER stripping)**

Table 1.2: Competitor Feature Comparison

2. System Requirements Specification

2.1 User Characteristics and Personas

- **Senior User (The Patient):** Age 60+. Low technical literacy, desires zero effort.
- **Clinician (The Doctor):** Neurologists who need data visualized instantly.
- **Caregiver (Family Member):** Requires automated push notifications for critical alerts.
- **IT Administrator:** Responsible for HIPAA compliance and system uptime.

2.2 User Stories

ID	User Story Description
US1	As a Senior, I want to connect my Facebook account once, so the system can monitor me without daily tests.
US2	As a Clinician, I want to view a 5-year trend graph of cognitive health, so I can intervene early.
US3	As a Caregiver, I want to receive an automated alert if my parent's risk score drops, so I can call a doctor.
US4	As an IT Admin, I want to ensure all text is anonymized immediately, so the hospital remains legally compliant.
US5	As a System Timer, I want to trigger a data sync every 24 hours to keep records up to date.

2.3 Functional Requirements (FRQ)

These requirements dictate the specific, testable behaviors the system must perform.

ID	Functional Requirement Specification
FRQ1	**Data Ingestion:** The system shall authenticate via OAuth 2.0 and retrieve the last 24 hours of text logs every day at 00:00 UTC.
FRQ2	**Reporting Module:** The system shall automatically generate downloadable PDF reports summarizing a patient's 5-year cognitive risk trend.
FRQ3	**AI Analysis:** The system shall generate a numeric Cognitive Risk Score (0.0 to 100.0) for every 100 words processed using ClinicalBERT.
FRQ4	**Automated Alerting:** The system shall generate a push notification to the Caregiver if a risk score drops by more than 15% over a 30-day average.
FRQ5	**Account Management:** The system shall allow users to securely create, update, and revoke API access to their linked profiles.
FRQ6	**Data Anonymization:** The system shall execute an NER algorithm to completely strip all names, locations, and dates before saving to the database.

2.4 Non-Functional Requirements (NFR)

These requirements dictate the quality attributes, performance, and security constraints of the system architecture.

ID	Non-Functional Requirement Specification
NFR1	**Performance:** The Clinician dashboard shall render a 5-year graph in under 3.0s using Redis.
NFR2	**Security:** All databases shall comply with HIPAA by encrypting data at rest (AES-256) and anonymizing PII within 2 minutes of ingestion.
NFR3	**Scalability:** Pipeline must process 10,000 text logs within a strict 60-minute batch window.
NFR4	**Reliability:** System architecture shall maintain 99.9% monthly uptime.
NFR5	**Usability:** The UI must comply with WCAG 2.1 AA accessibility (high contrast, large fonts).
NFR6	**Maintainability:** The backend must utilize a modular microservices architecture.

2.5 Requirements Traceability Matrix

Requirement	US1 (Senior)	US2 (Doc)	US3 (Family)	US4 (Admin)	US5 (Timer)
FRQ1 (Ingest)	X				X
FRQ2 (Reports)		X			
FRQ3 (Score)		X			X
FRQ4 (Alert)			X		
FRQ5 (Accounts)	X			X	
FRQ6 (Anonymize)				X	X

3. System Components and Detailed Use Cases

This chapter provides a deep dive into each system component. Every use case is detailed with its specific flow, use case diagram, explanation, sequence diagram, and (where applicable) state diagram.

3.1 UC 1: Account Management and Consent

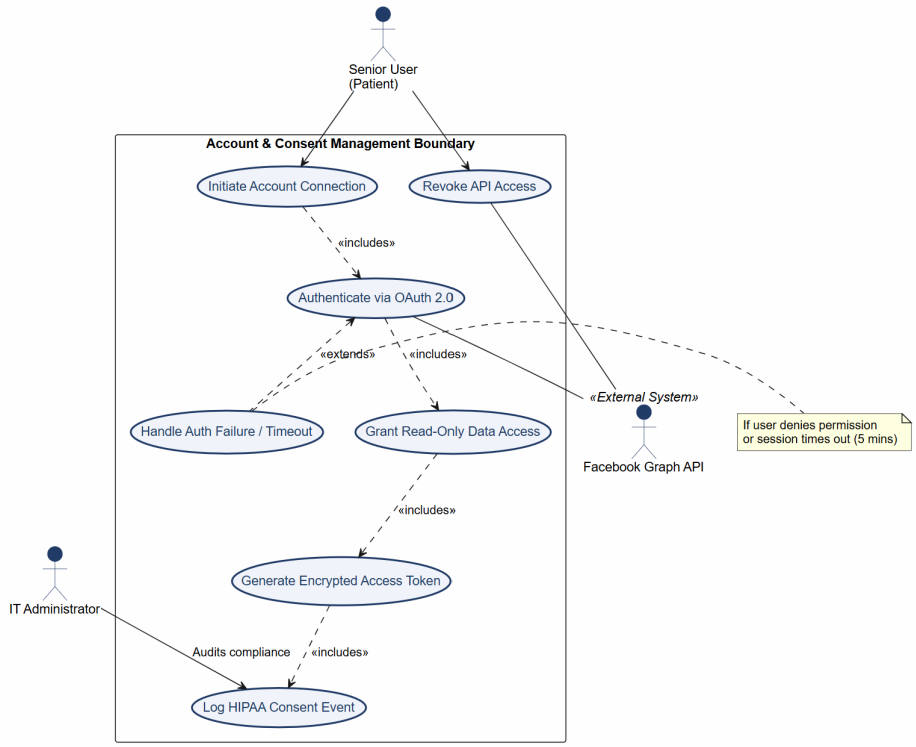


Figure 3.1: Use Case Diagram: Account Management.

Explanation: The Senior User initiates the system by securely authorizing Facebook via OAuth. No passwords are ever stored locally.

Use Case 1 Specification	
Description	The Senior User authenticates via Facebook OAuth to grant read-only access.
Inputs	User clicks OAuth link, inputs social media credentials.
Processing	Validates credentials and generates an encrypted access token.
Outputs	Secure token saved to Database; initial 5-year historical sync begins.
Constraints	Must comply with OAuth 2.0 security protocols and timeout after 5 minutes.

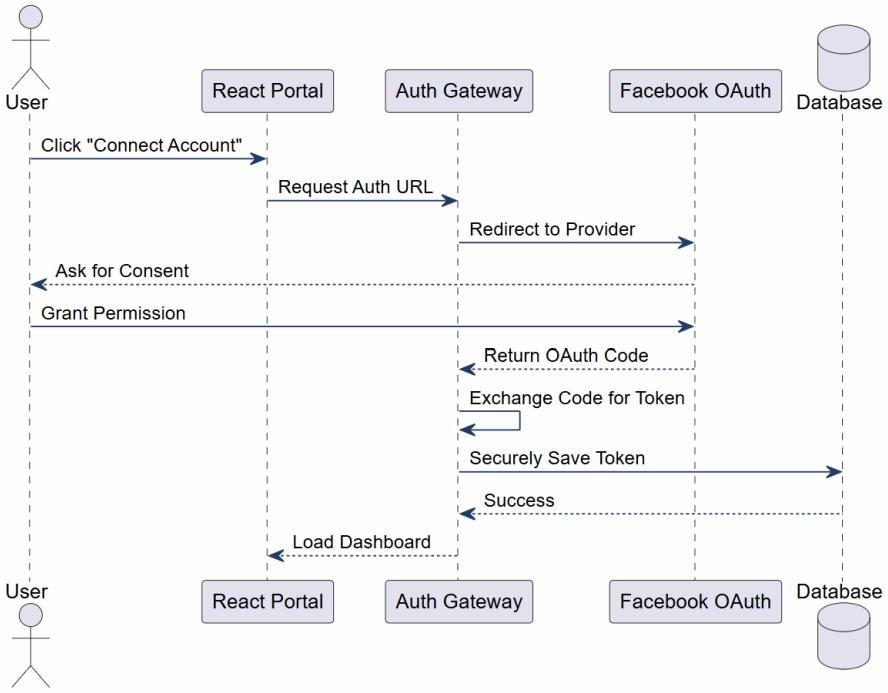


Figure 3.2: Sequence Diagram: OAuth API Authorization Flow.

3.2 UC 2: Execute Daily Passive Sync

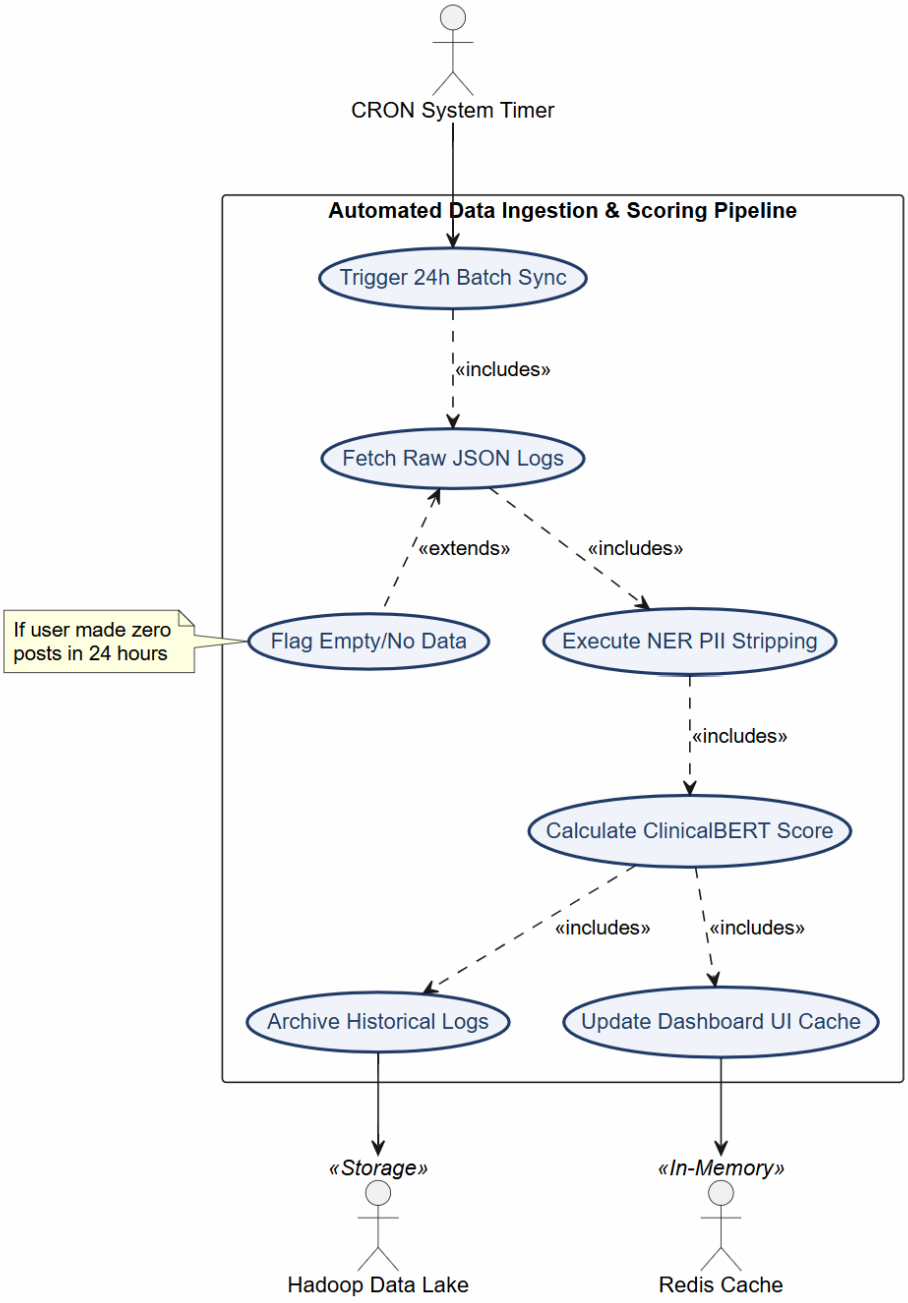


Figure 3.3: Use Case Diagram: Daily Batch Ingestion Pipeline.

Explanation: This is the core automated backend process. Triggered by a timer, it downloads the text, strips PII via NER to protect privacy, and scores the cognitive risk using the AI.

Use Case 2 Specification	
Description	The System Timer triggers a batch download, strips PII, and generates risk scores.
Inputs	24-hour internal timer trigger, Raw JSON text arrays from Facebook API.
Processing	Strips PII using NER, tokenizes text, and calculates cognitive risk using ClinicalBERT.
Outputs	Float variable (Risk Score) permanently saved to Hadoop Data Lake.
Constraints	10,000 logs must be processed within 60 minutes to prevent timeouts.

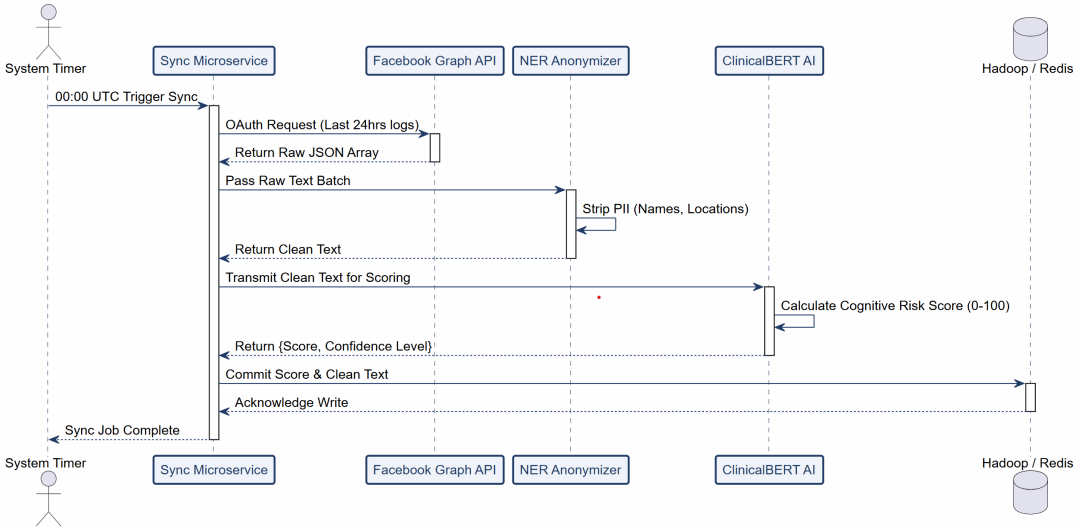


Figure 3.4: Sequence Diagram: Data Sync, Anonymization, and AI Scoring.

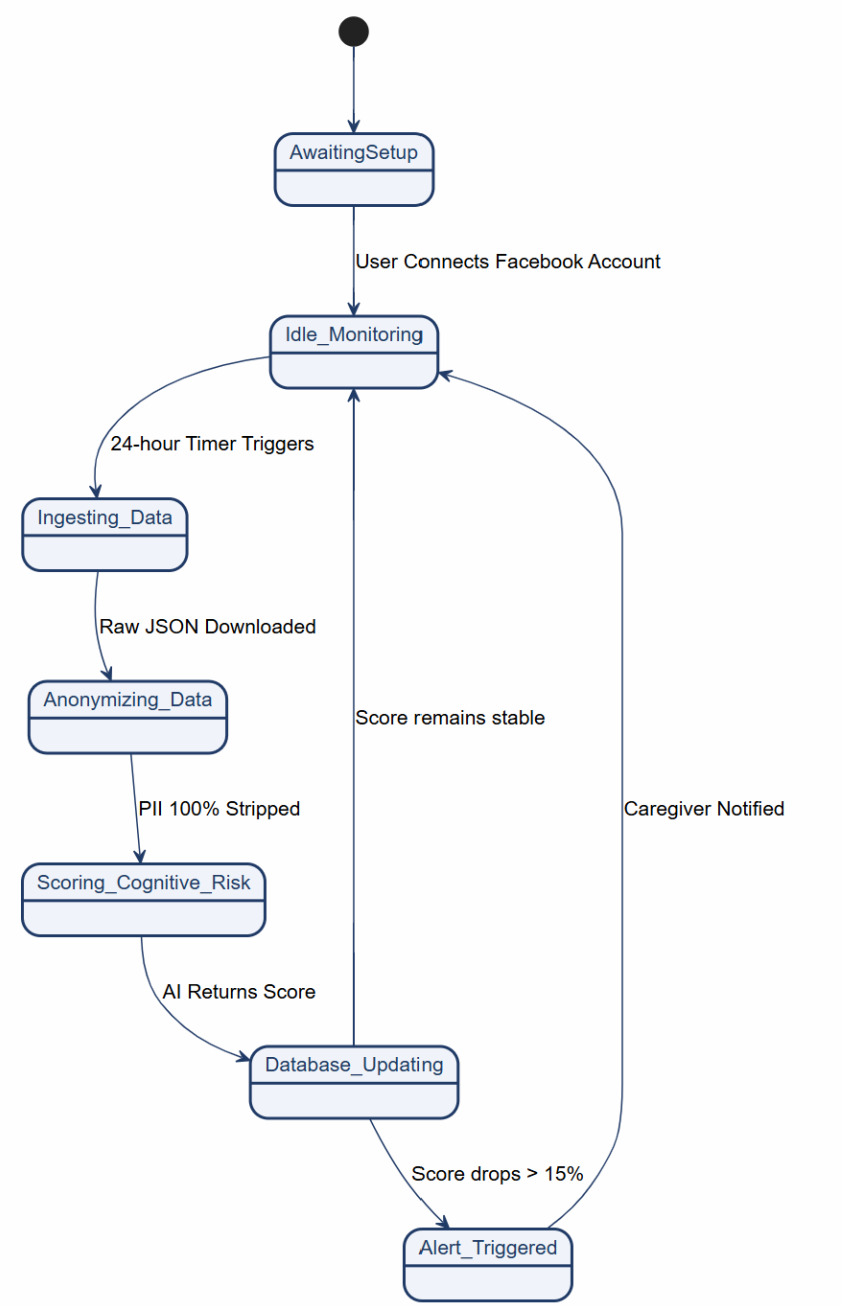


Figure 3.5: State Diagram: The 24-hour monitoring lifecycle.

3.3 UC 3: View Clinical Dashboard & Reports

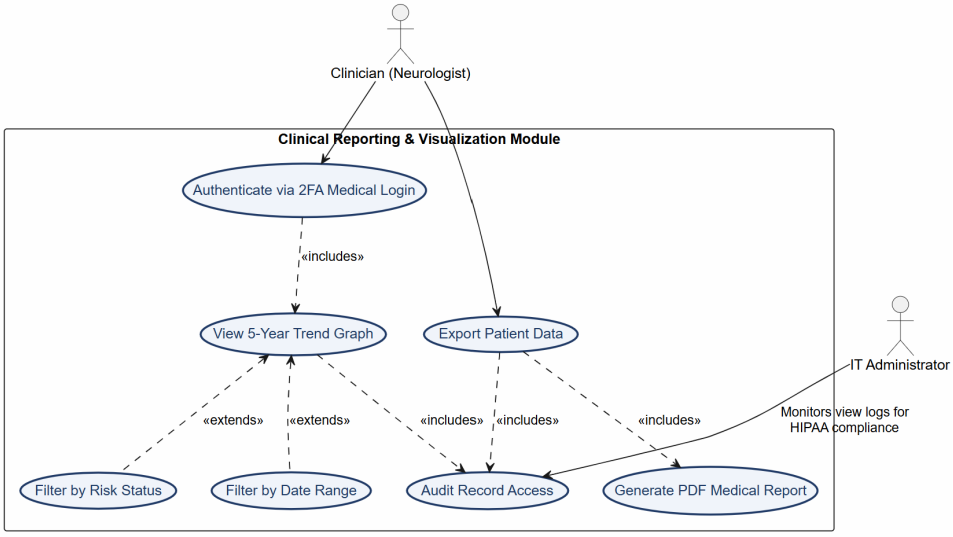


Figure 3.6: Use Case Diagram: Clinician Reporting Module.

Explanation: Clinicians access the front-end React interface to monitor their patients. They can view graphs or export formal PDF reports for medical files.

Use Case 3 Specification	
Description	The Clinician queries patient data and exports analytical reports.
Inputs	Clinician selects a specific patient ID on the dashboard.
Processing	API Gateway queries Redis cache for real-time graph rendering data.
Outputs	Interactive UI line graph and downloadable PDF report.
Constraints	Graph must render under 3.0 seconds (NFR1).

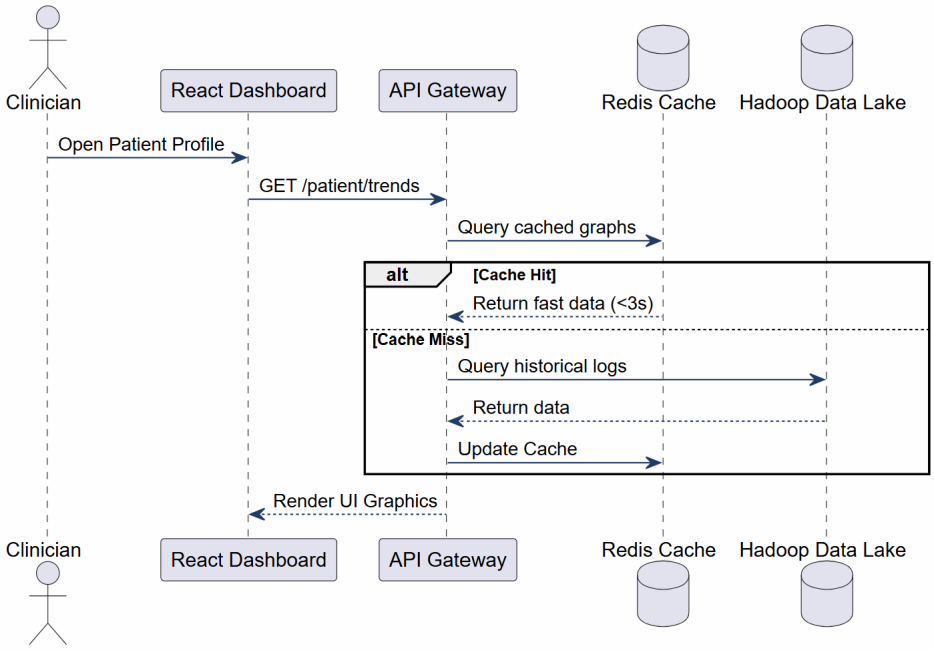


Figure 3.7: Sequence Diagram: Dashboard Query and Caching.

3.4 UC 4: Trigger Automated Risk Alerts

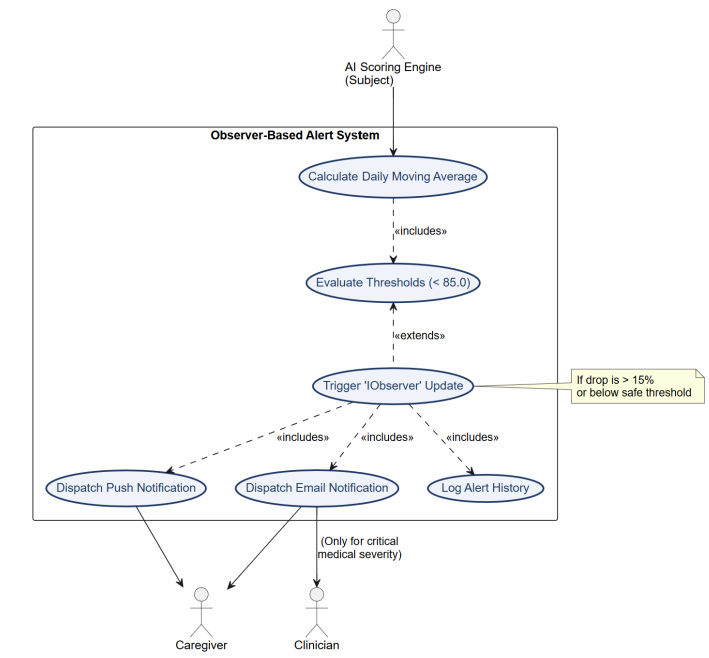


Figure 3.8: Use Case Diagram: Observer-based Alert System.

Explanation: Utilizing the Observer Design Pattern, the system constantly monitors the generated scores. If a critical drop is detected, it alerts family members instantly.

Use Case 4 Specification	
Description	The system automatically notifies a Caregiver if health declines rapidly.
Inputs	AI outputs a score drop ¿15% over a 30 day moving average.
Processing	The AlertDispatcher microservice formats a push notification.
Outputs	An Email/SMS sent to the linked Caregiver account.
Constraints	Must dispatch within 60 seconds of score calculation.

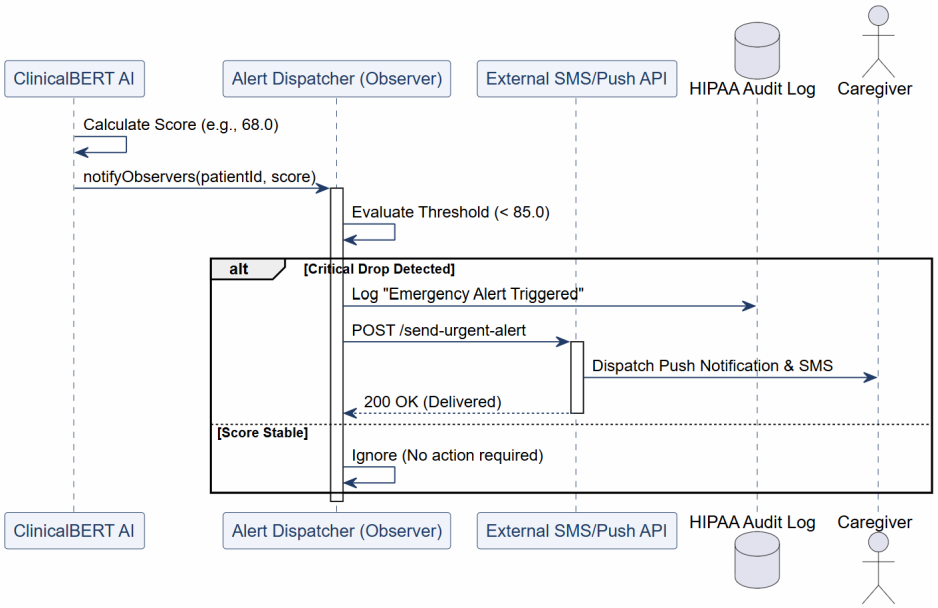


Figure 3.9: Sequence Diagram: Observer Pattern Triggering the Alert Dispatcher.

3.5 UC 5: Monitor System Security and Compliance

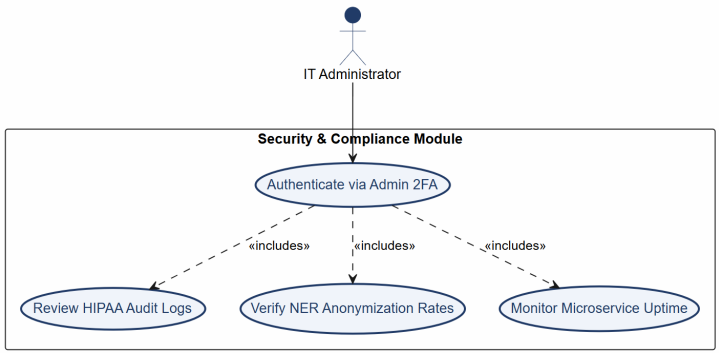


Figure 3.10: Use Case Diagram: Security and Compliance Module.

Explanation: The IT Administrator accesses the backend to monitor system health and ensure strict compliance with HIPAA data regulations by reviewing audit logs.

Use Case 5 Specification	
Description	IT Admin monitors HIPAA audit logs and verifies system anonymization success.
Inputs	Admin inputs 2FA credentials to access the security portal.
Processing	System queries the HIPAA Audit Database for access records.
Outputs	A dashboard displaying compliance metrics and microservice uptime.
Constraints	Admin sessions must automatically terminate after 15 minutes of inactivity.

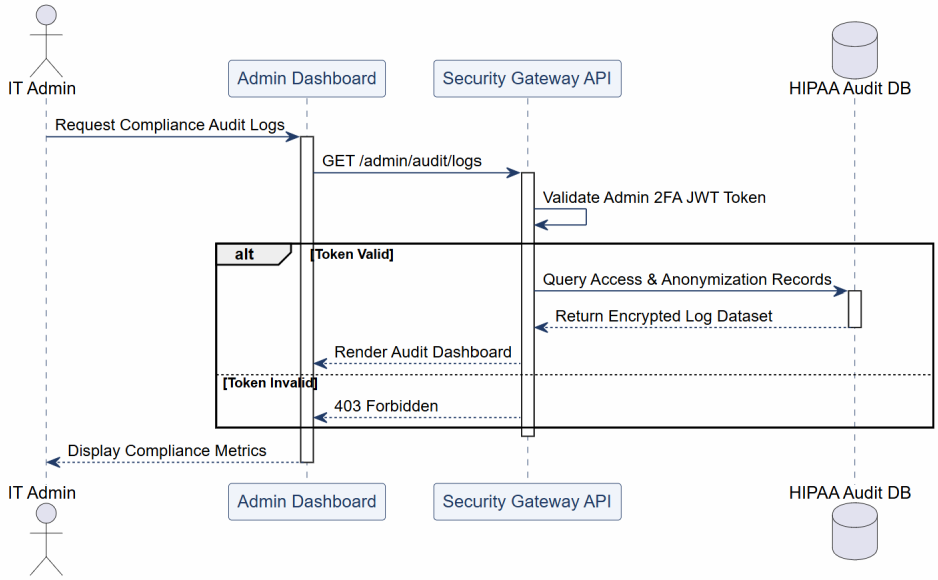


Figure 3.11: Sequence Diagram: Admin querying the secure HIPAA Audit Database.

4. System Architecture Models

4.1 System Context Diagram (C4 Level 1)

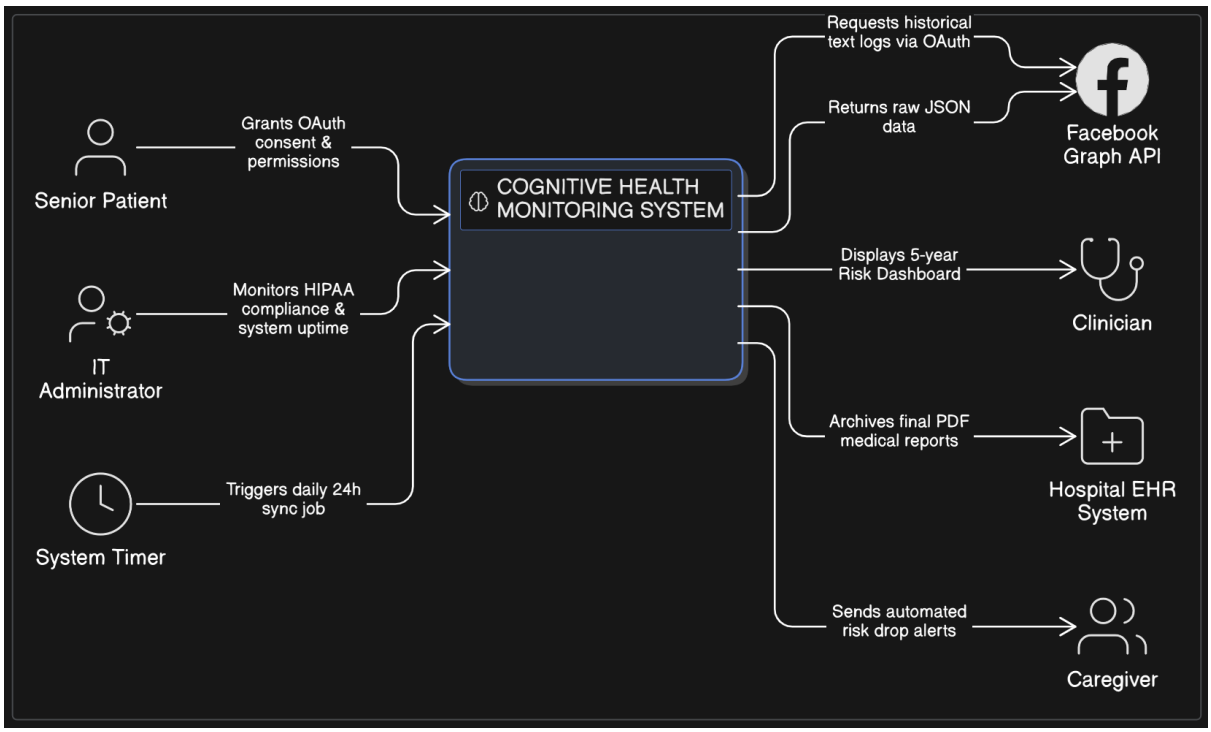


Figure 4.1: Level 1 Context Diagram highlighting external system dependencies.

4.2 Container Diagram (C4 Level 2)

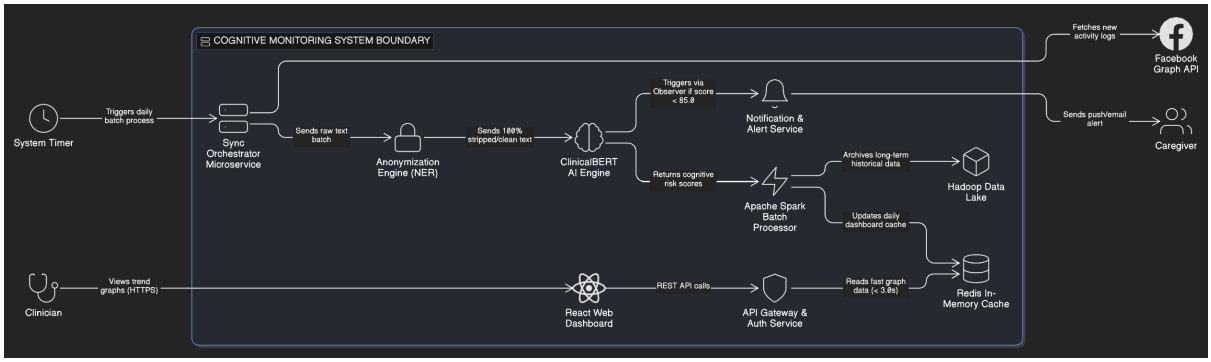


Figure 4.2: Level 2 Container Diagram showing internal Spark and Hadoop microservices.

4.3 Activity Diagram (Logic Flow)

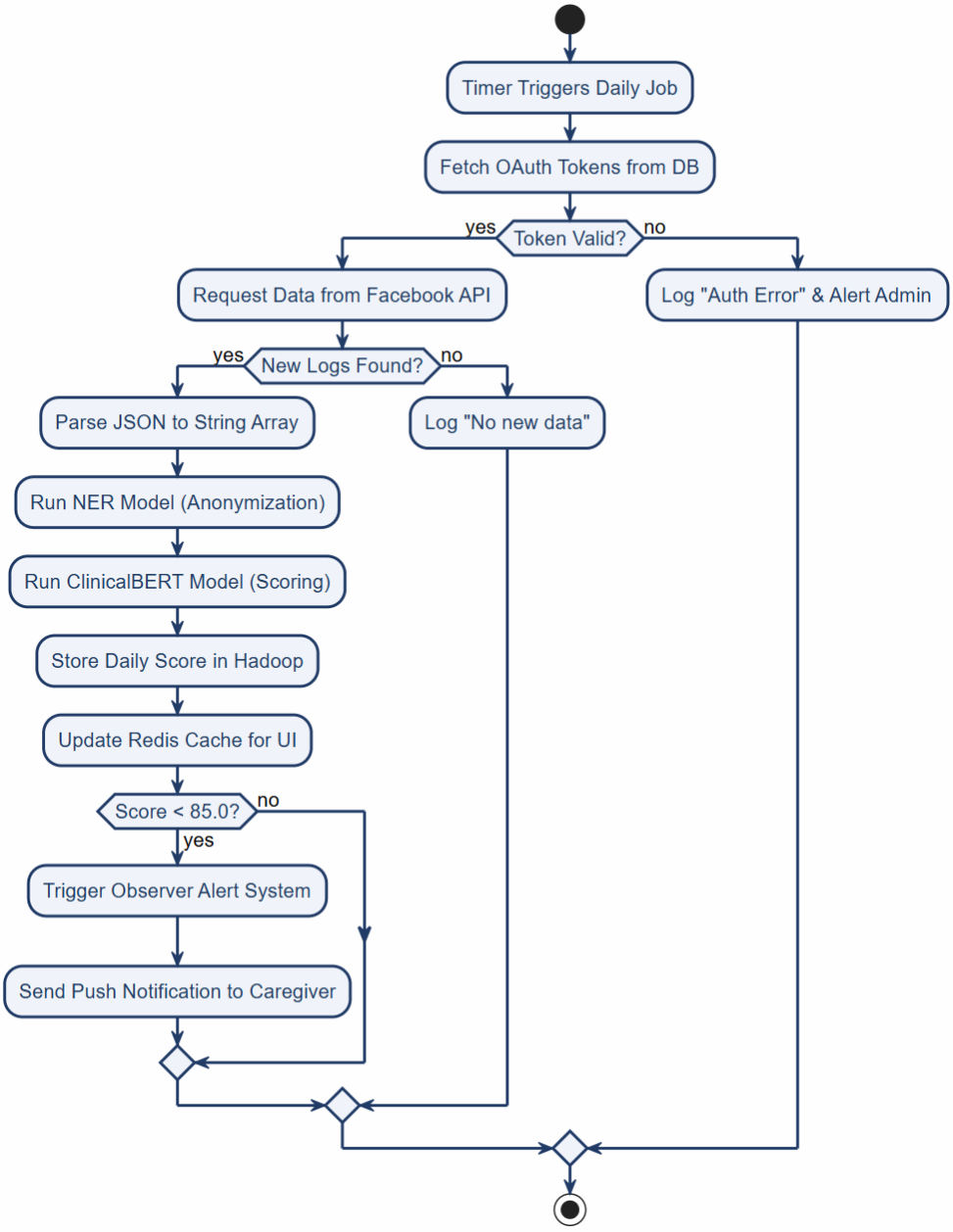


Figure 4.3: Activity Diagram mapping the conditional logic flow of the sync engine.

4.4 UML Class Diagram

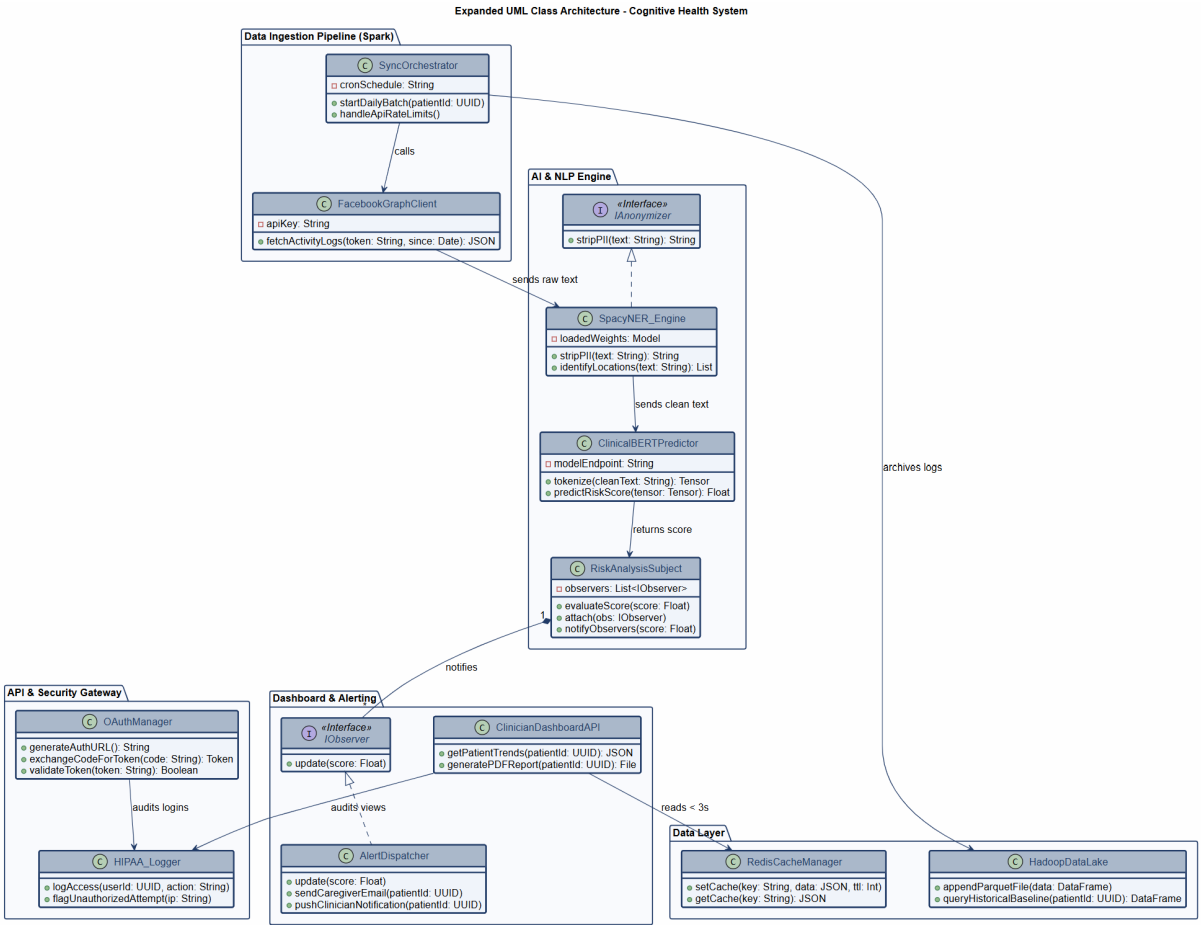


Figure 4.4: UML Class Diagram highlighting Object-Oriented packages and the Observer Pattern.

4.5 Entity Relationship (ER) Diagram

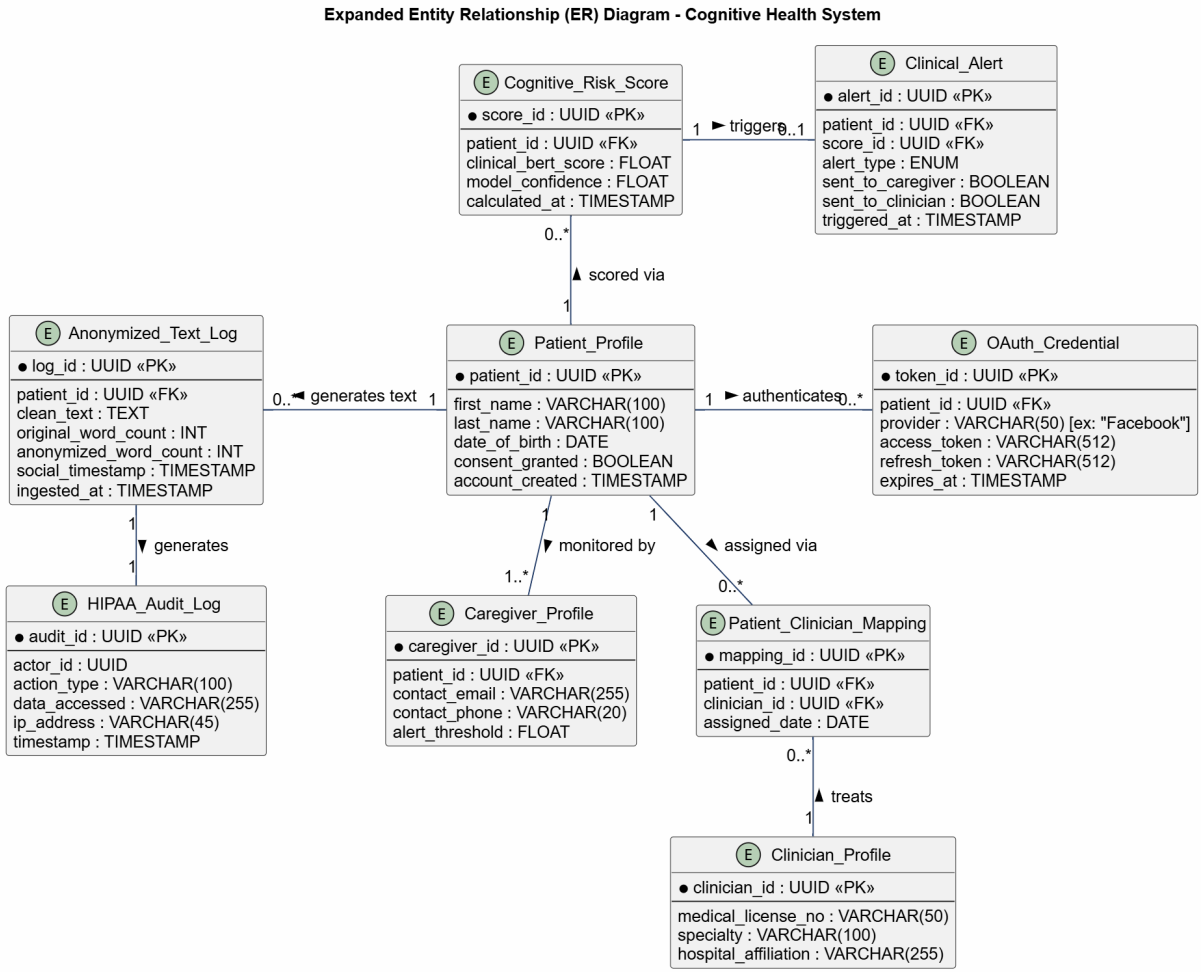


Figure 4.5: Expanded Entity Relationship Database Schema for HIPAA Compliance.

5. Validation and Interfaces

5.1 External Interfaces & UI/UX Prototype

The interface is designed with a responsive, multi-platform approach to serve different user personas. The **Clinician Dashboard (Desktop)** provides dense data visualizations for deep medical analysis, while the **Caregiver App (Mobile)** focuses on immediate alerts, large tap targets, and rapid communication capabilities.

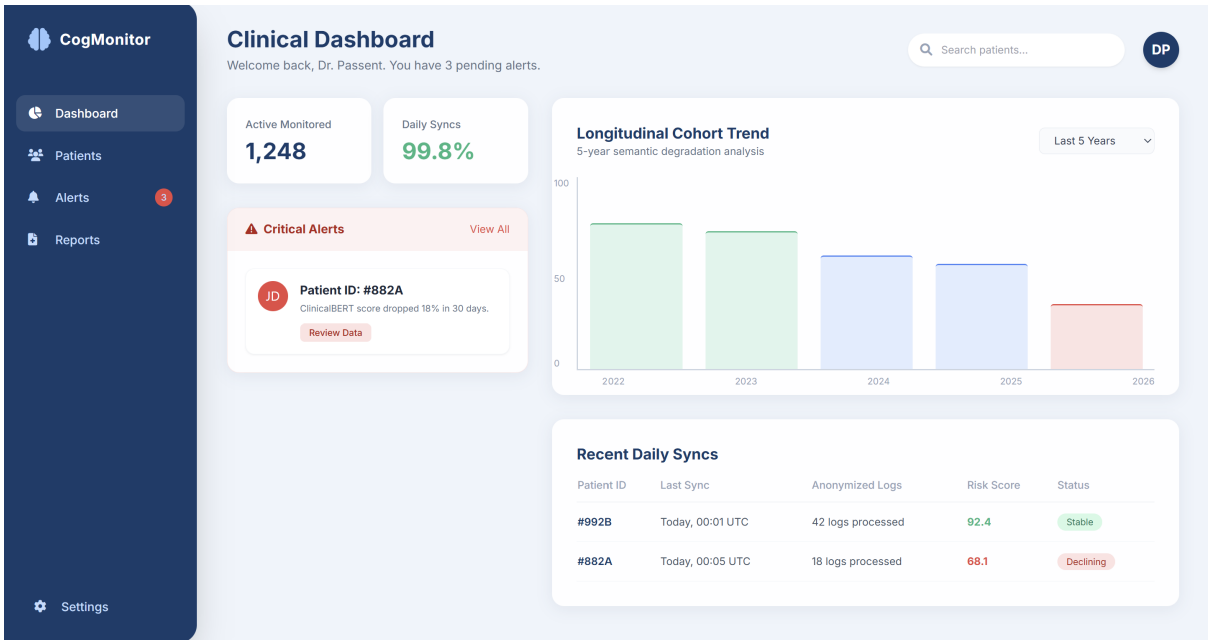


Figure 5.1: UI/UX Prototype Wireframe for the Clinician Dashboard (Desktop).

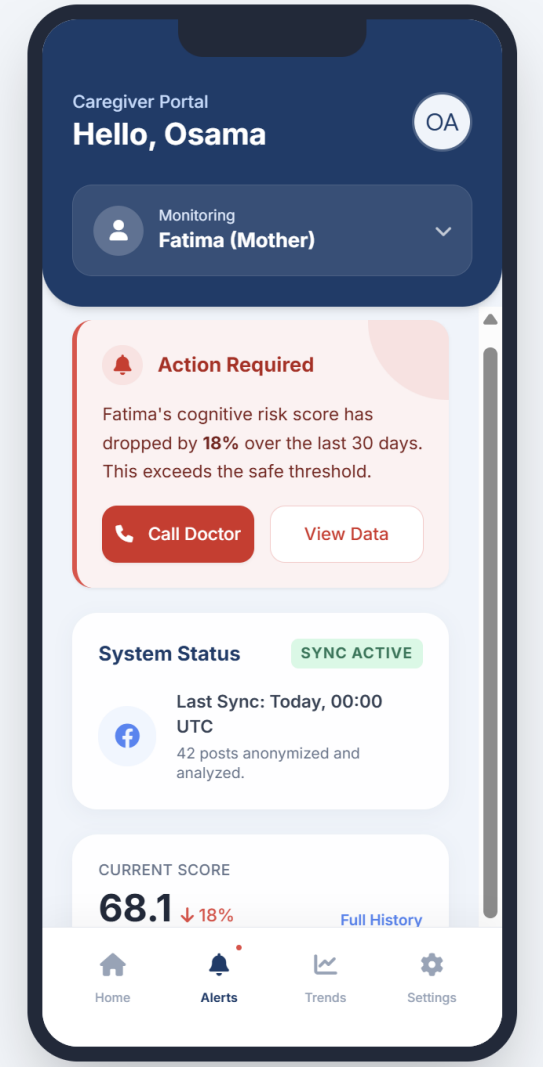


Figure 5.2: UI/UX Prototype for the Caregiver Companion App (Mobile View), highlighting the critical alert system triggered by a score drop.

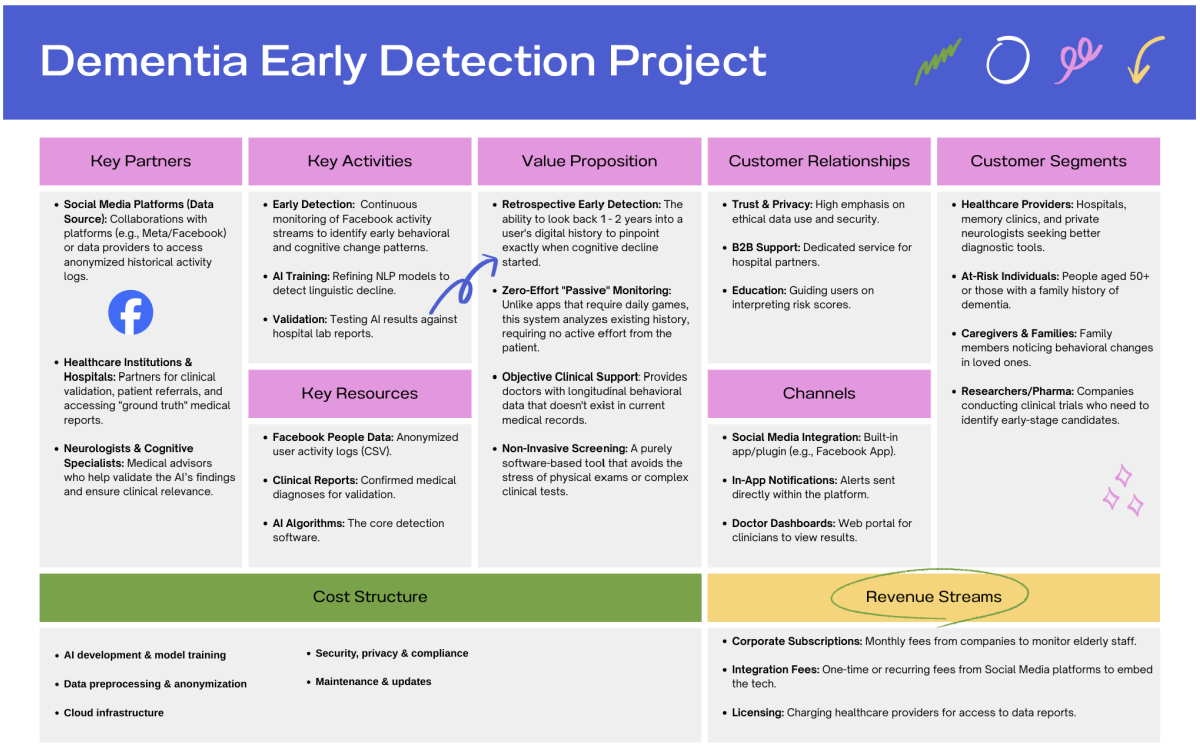
5.2 Validation and Acceptance Criteria

Target Requirement	Acceptance Test Condition
Account Auth (FRQ1)	A test user must successfully connect an account and receive a token within 10 seconds.
Anonymization (FRQ6)	Fed a file of 500 fake names, the NER engine must output text with 100% redaction.
Rendering (NFR1)	System must load a graph containing 1,825 data points (5 years) in under 3.0 seconds.
Alerting (FRQ4)	Injecting a score drop of 20% must result in an email sent to the Caregiver within 60 seconds.

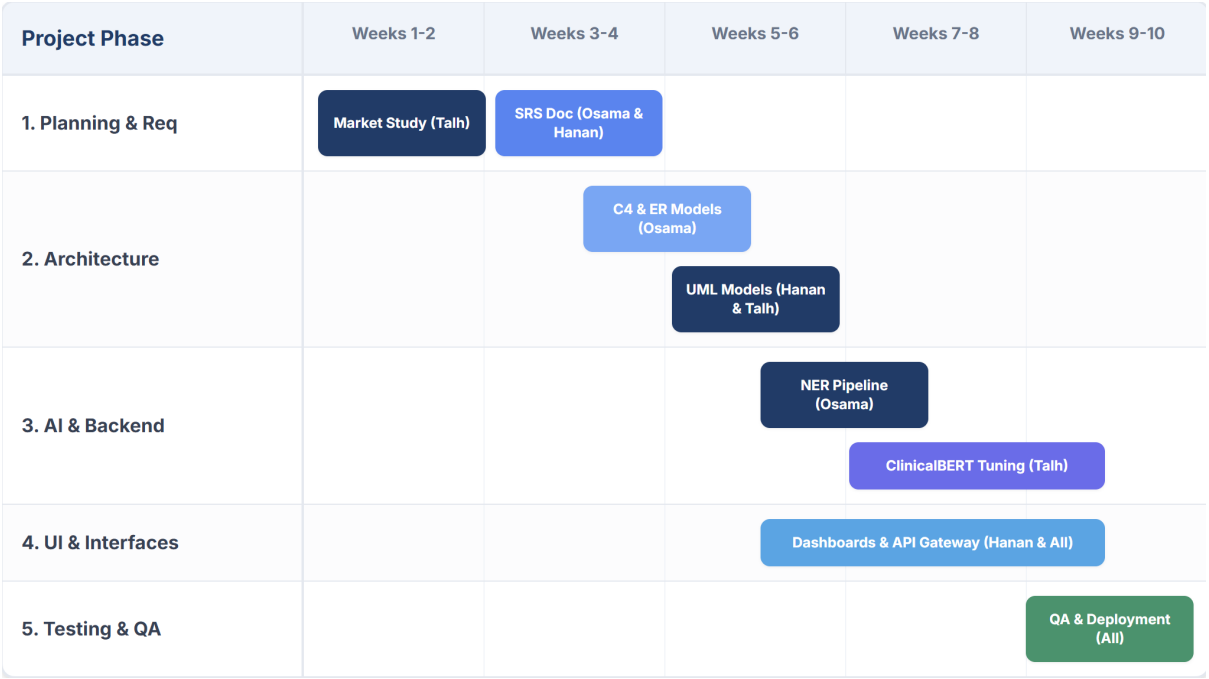
Table 5.1: System Acceptance Criteria

6. Appendices

6.1 Appendix A: Business Model Canvas



6.2 Appendix B: Project Gantt Chart



7. References

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